How Much of a Probe Direction Lives in the Input Embedding? An Audit of Four Published Claims

Anonymous ACL submission

Abstract

Interpretability research in language models often identifies latent directions in the hidden state that track specific behaviors, such as truthfulness, refusal, sentiment, or sycophancy. The usual evidence is a high correlation between projecting onto that direction and the behavior. The probing literature already provides a matched control by isolating out a baseline representation, and the single-direction line of work has not adopted this approach. We apply that control to four published direction claims across a panel of open-weight chat models, using the input-embedding projection as the baseline. On a harmful-prompt benchmark in two specific checkpoints, a published refusal direction loses 63–66% of its predictive correlation under the control, even though projecting the direction out of the network still suppresses refusals, a dissociation that does not appear on four held-out chat families.

1 Introduction

When a language model refuses a harmful request, asserts a factual claim, or expresses a sentiment, the choice is computed inside the network from a pattern of numerical activations. A recurring claim in the interpretability literature is that this pattern can be reduced, for each behavior, to a single direction in the model's hidden representation that lines up with the behavior (Park et al., 2024). Names attached to such directions include the truth direction (Marks and Tegmark, 2024), the refusal direction (Arditi et al., 2024), the sentiment direction (Tigges et al., 2024), and the sycophancy direction (Panickssery et al., 2024). The evidence behind each claim takes one of two forms. A small classifier fit on the projection of the hidden state onto the candidate direction is found to correlate with the behavior (a procedure called *linear probing*), or the behavior shifts when the direction is subtracted from the network's activations (a procedure called *directional ablation*).

Both evidences treat the direction as something the network builds through depth. However, neither is held out against an alternative. The very first layer of a transformer (the input-embedding layer, where each token becomes a fixed vector) already produces a representation in which projecting onto a candidate direction can correlate with the behavior, before any transformer block runs. A high probe correlation at a middle layer can therefore be inherited from this input-embedding projection rather than constructed by the model. Conditional probing (Hewitt et al., 2021) addresses this kind of confound for classification probes by partialling out a baseline representation, and the single-direction interpretability line of work has not adopted the analogous control for direction projections.

To diagnose this gap, we test whether a high read-layer correlation reflects structure the model builds or structure inherited from the input embedding. The audit pairs a held-out partial-correlation control, which subtracts the embedding-layer projection from the read-layer projection, with a random-direction null and a behavioral intervention that subtracts the direction from the activations. On widely cited truth and sentiment directions, the control passes silently. On a published refusal direction in two specific chat models on a single harmful-prompt benchmark, it flags a real input-inherited contribution and reduces the correlation to within the random-direction null. Yet on the same two cells, subtracting the direction from the activations still suppresses refusals on held-out prompts (Figure 2). Predictive correlation and behavioral intervention dissociate on those cells. A fourth case, sycophancy, fails differently. The read-layer probe correlation is indistinguishable from zero, so no learned-direction claim is left to audit. The main contributions of the study are as follows.

(i) We specialize conditional probing (Hewitt et al., 2021) to scalar difference-of-means direction claims, quantifying how much of a probe correlation comes from the input embedding rather than from the model's later computation.

(ii) We audit several published direction claims on a panel of open-weight chat models, with positive and negative findings reported together.

(iii) We follow up interventionally on the cells that fail the predictive control, showing that predictive evidence and behavioral evidence can come apart on the same direction.

2 Related Work

Validity controls for linear probes. A long line of work asks whether a probe is reading a model's internal representation or exploiting an artifact. Random-label baselines and information-theoretic probes bound how much accuracy a probe can reach without genuine representational content (Hewitt and Liang, 2019; Pimentel et al., 2020; Voita and Titov, 2020; Belinkov, 2022). The closest methodological precedent is conditional probing (Hewitt et al., 2021), which partials out a baseline representation, typically non-contextual word embeddings, before measuring what additional information the read layer carries. Our control is the linear, scalar specialization of that idea to single-direction claims, with the input-embedding projection along the candidate direction taking the role of the word-identity baseline. The within-model embedding-inheritance concern we measure is a close analogue of the cross-model textual-leakage concern raised by Boxo et al. (2025).

Concept erasure and amnesic methods. Amnesic probing (Elazar et al., 2021) and concept-erasure methods such as iterative nullspace projection (Ravfogel et al., 2022) and least-squares concept erasure (Belrose et al., 2023) modify the representation to remove a target concept and then read off the downstream effect. Our control leaves the representation untouched. It asks whether a specific candidate direction adds predictive value beyond what the input embedding already provides along the same axis. The two approaches probe complementary aspects of representational validity.

Single-direction claims and directional ablation. The truth (Marks and Tegmark, 2024), refusal (Arditi et al., 2024), sentiment (Tigges et al., 2024), and sycophancy directions from contrastive activation addition (CAA, Panickssery et al., 2024) are the immediate targets of our audit. The interventional half of the audit uses the directional-ablation procedure of Arditi et al. (2024), which builds on the steering and representation-engineering tradition (Zou et al., 2023a; Turner et al., 2023).

Causal abstraction, geometry, and recent validity work. Activation-patching and causal-abstraction work scores the localization of a behavior through hidden-state interventions (Geiger et al., 2025; Zhang and Nanda, 2023; Jin and Rinard, 2024), and recent work tightens the validity standard for single-direction claims with geometric refinements (Wollschläger et al., 2025; Canby et al., 2025) and refusal-axis reliability studies (Yu et al., 2025). The embedding-inheritance confound sits among these without being addressed by any of them.

3 Method

3.1 Dataset

We use four public datasets. **AdvBench** (Zou et al., 2023b) contains 520 harmful behavior instructions, from which we sample 250 (seed 0). **Alpaca** (Taori et al., 2023) contains 52K harmless instructions. We keep no-input items with 20–150 characters and sample 250 (seed 0). **Marks–Tegmark Cities** (Marks and Tegmark, 2024) contains 1,496 "The city of X is in Y" statements with binary truth labels. We sample 250 true and 250 false statements (seed 0). **Conjunction-fallacy suite** (this paper) contains 150 prompt triples $(A, B, A \wedge B)$ across canonical Linda-style items (Tversky and Kahneman, 1983; Wang et al., 2024; Macmillan-Scott and Musolesi, 2024), paraphrase variants, concept-combination items, and disjunctions. The κ-direction audit on this suite is reported in Appendix F. Representative items from each dataset appear in Appendix G.

3.2 Models

Our main analysis uses seven open-weight instruction-tuned chat LLMs with 3.8B to 72B parameters, drawn as a convenience sample from five families with public Hugging Face releases as of April 2026. The models include Llama-3-8B-Instruct (Meta AI, 2024a), Llama-3.1-70B-Instruct (Meta AI, 2024b), Mistral-7B-Instruct-v0.3 (Mistral AI, 2024), Qwen2.5-7B-Instruct, Qwen2.5-72B-Instruct (Qwen Team, 2024), Gemma-2-9B-it (Google DeepMind, 2024), and Phi-3-mini-4k-instruct (Microsoft, 2024). The detailed information about the models are included in Appendix C. Additional Qwen sizes used in the conjunction-fallacy scaling sweep appear in Appendix D. The held-out-family replication (Appendix E) uses OLMo-2, Yi-1.5, Falcon-3, and Tulu-3, four families disjoint from the main-analysis set.

3.3 Behavior outcomes and direction fitting

Each audited direction supplies a per-prompt behavioral outcome y_i and a read-layer direction $\hat{d}_\ell$ fit on a training fold. The held-out evaluation then correlates the projection on $\hat{d}_\ell$ with y_i.

Mean-difference directions. Refusal and truth use the same direction-fitting form. In both cases, $\hat{d}_\ell$ is the unit-normalized difference between the class-conditional mean activations at the read layer. The two audits differ in the labels used to fit the direction and in the behavioral outcome used for evaluation.

Refusal direction. The behavioral outcome is refusal log-odds, computed teacher-forced at the final prompt token. Eq. 1 shows an example calculation of log-odds of prefixes. The robustness panel in §4.4 also reports a rule-based variant that classifies a generation as a refusal when its first 32 tokens contain any of the canonical refusal prefixes used by Arditi et al. (2024) ("I cannot", "I can't", "I am unable", "I refuse", and variants). We use a train fold of 200 harmful (AdvBench) and 200 harmless (Alpaca) prompts. The held-out test fold contains 50 harmful and 50 harmless prompts ($n_{\text{test}} = 100$). The direction $\hat{d}_\ell$ is the unit-normalized mean-difference $\bar{h}_{\text{harmful}} - \bar{h}_{\text{harmless}}$ at the read layer, following Arditi et al. (2024). We use L_{24} for Llama-3-8B-Instruct and Mistral-7B-Instruct-v0.3, and L_{36} for Gemma-2-9B-it.

$$y_{\text{ref}} = \log \frac{P(\text{“ I cannot help”} \mid p)}{P(\text{“ Sure, here is”} \mid p)}, \qquad (1)$$

Truth direction. The behavioral outcome is the true–false elicitation score (Eq. 2), where p_e is a fixed elicitation prompt appended to statement p. This readout is a behavioral analogue of, but not identical to, Marks and Tegmark (2024)'s direct hidden-state probe accuracy. The train fold contains 200 true and 200 false Cities statements. The test fold contains 50 true and 50 false statements. The direction protocol is identical to refusal, using a unit-normalized mean-difference direction with binary truth labels at the same read layers.

$$y_{\text{tr}} = P(\text{“ True”} \mid p_e) - P(\text{“ False”} \mid p_e), \qquad (2)$$

The conjunction-fallacy direction κ uses a different fitting protocol because each item is a prompt triple rather than a binary-labeled prompt, and is reported entirely in Appendix F.

3.4 The partial-correlation control

The control is the linear, scalar specialization of conditional probing (Hewitt et al., 2021) to single-direction claims, with the projection scalar replacing classification-probe accuracy and the input-embedding projection along the candidate direction replacing the non-contextual word-embedding baseline.

Let $h_{i,\ell} \in \mathbb{R}^d$ denote the residual-stream hidden state of prompt i at layer ℓ, y_i the per-prompt behavioral outcome defined in §3.3, and $\hat{d}_\ell$ the direction-of-interest fit on a training fold by the original protocol (mean-difference, PCA, or ridge regression), which are unit-normalized. The standard reported quantity in linear-probe interpretability is the read-layer probe correlation (Eq. 3).

A matching control is introduced at the input-embedding layer L_0. We re-fit a direction $\hat{d}_0$ at L_0 using the same training protocol and compute the L_0 *correlation* (Eq. 4) against mean-pooled activations $\bar{h}_{i,0}$, which are a mask-weighted bag of input embeddings. This value is a model-agnostic proxy for how much of any read-layer correlation could be inherited from the token-level input representation.

The partial correlation (Eq. 5) then removes the shared variance attributable to the L_0 projection, where $r_{\ell,0}$ is the Pearson correlation between the two projection series.

$$r_\ell = \operatorname{corr}\left(h_{i,\ell} \cdot \hat{d}_\ell,\ y_i\right). \qquad (3)$$

$$r_0 = \operatorname{corr}\left(\bar{h}_{i,0} \cdot \hat{d}_0,\ y_i\right) \qquad (4)$$

$$r_{\text{partial}} = \frac{r_\ell - r_0\, r_{\ell,0}}{\sqrt{(1 - r_0^2)(1 - r_{\ell,0}^2)}} \qquad (5)$$

For each (direction, model) cell, we report (a) raw r_ℓ, (b) embedding-layer r_0, (c) r_{partial}, (d) drop-percent $= 100 \times (1 - r_{\text{partial}}/r_\ell)$, and (e) 95% percentile bootstrap confidence intervals from 2,000 resamples on the held-out test fold.

Verdict legend. We assign a categorical verdict to each cell. A cell is **DEGENERATE** if the raw-r_ℓ bootstrap CI includes zero, which means there is no learned-direction correlation to audit. Among non-degenerate cells, **PASS** requires $|r_{\text{partial}}| \geq 0.7\,|r_\ell|$. **PASS-b** (borderline pass) requires $0.5\,|r_\ell| \leq |r_{\text{partial}}| < 0.7\,|r_\ell|$. **BORDERLINE** indicates that an alternative decision rule shifts the cell across the PASS-b threshold. **FAIL** applies otherwise. We additionally flag *dissociation* cells (§4.4) where the partial-correlation verdict and the directional-ablation effect (Eq. 6) disagree. The 0.7 and 0.5 thresholds correspond to approximately 30% and 50% reductions in apparent predictive content, consistent with standard power-reduction magnitudes (Cohen, 1988). The DEGENERATE category was introduced inductively after observing the sycophancy null (§4.7), and the rule itself is data-independent.

Caveats. Four caveats bound the control. First, fitting both $\hat{d}_\ell$ and $\hat{d}_0$ within the same training fold could bias r_{partial} downward due to shared label overfitting. We address this with a 20-fold disjoint-split refit procedure in §4.5. Second, the control compares mean-pooled L_0 activations with final-token L_{read} activations and corrects only the linear share of the embedding contribution (Appendix H). The operational protocol is to report both mean-pool and last-token L_0 whenever $n_{\text{test}} \lesssim 100$ (Appendix F shows one flip at $n = 30$). Third, any non-linear contribution from the input embedding to the behavioral outcome is not removed by partialling out the linear projection, which biases r_{partial} upward toward PASS. Fourth, the control is complementary to random-label (Hewitt and Liang, 2019), amnesic (Elazar et al., 2021), and textual-leakage (Boxo et al., 2025) controls (Jin and Rinard, 2024), not a substitute.

Appendix H relates Eq. 5 to mediation-analysis (Pearl, 2009) and concept-erasure controls (Elazar et al., 2021; Belrose et al., 2023).

3.5 Statistical procedures

We use percentile bootstrap with $n_{\text{boot}} = 2000$ resamples and fixed seed (`seed=0`, unstratified). Holm–Bonferroni correction is applied to bootstrap interval tails over the 14-test refusal block and the full 29-cell panel. The correction targets the tail-test family, not the categorical verdicts themselves. The 0.7/0.5 verdict thresholds are heuristic, and we check verdict stability under relaxing PASS-b from 0.7 to 0.5. The random-direction null of §4.5 uses $n_{\text{rand}} = 200$, giving a binomial 95% CI on percentile rank of ±3–5 pp. Appendix H discusses stratified-bootstrap and larger-n_{rand} alternatives. Compute requirements are reported in Appendix I.

4 Results

For each (direction, model) cell, the audit fits the direction at the read layer and at the embedding layer, computes the partial correlation between their projections and the behavioral outcome, and assigns a verdict from bootstrap intervals on raw r, partial r, and the drop percentage. Figure 1 sketches the audit pipeline that we apply to every cell. Figure 2 highlights the truth-calibration and refusal cells. Table 1 consolidates the full audit across four directions and seven models. Per-direction details follow in Sections 4.1 through 4.5.

4.1 Truth direction, three of three pass

The truth direction passes on all three audited models. Partial r matches raw r within bootstrap noise against the elicitation outcome of Eq. 2, and the drop sits in $[-1\%, +4\%]$ across the three cells. The mean-pooled embedding-layer projection is anti-correlated with the truth label on every model, which means the input embedding does not encode truthfulness and the network builds the readout through depth. When a learned direction is genuine, the audit passes silently.

4.2 Sentiment direction, six of seven pass

The Tigges-2023 sentiment direction (Tigges et al., 2024) is fit on SST-2 (Socher et al., 2013) with 250 positive and 250 negative items, an $80/20$ train/test split, a mean-difference protocol, and a 2000-resample bootstrap. Six of seven chat LLMs (Llama-3-8B, Llama-3.1-70B, Mistral-7B-v0.3, Gemma-2-9B-it, Qwen2.5-7B, Qwen2.5-72B) return a clean PASS. Partial r runs from $+0.52$ to $+0.79$ (raw r up to $+0.81$), drops sit between 3% and 14%, and the partial-r bootstrap interval is strictly positive on every passing cell with a lower bound at or above $+0.37$. The embedding-layer projection carries some sentiment signal, with r_0 in $[+0.32, +0.44]$, as expected for a content property, yet the network adds substantial information and partial r tracks raw r within 14% on every passing cell. Phi-3-mini is excluded from this panel by a transformers-library configuration error (`KeyError: 'type'`), and this does not

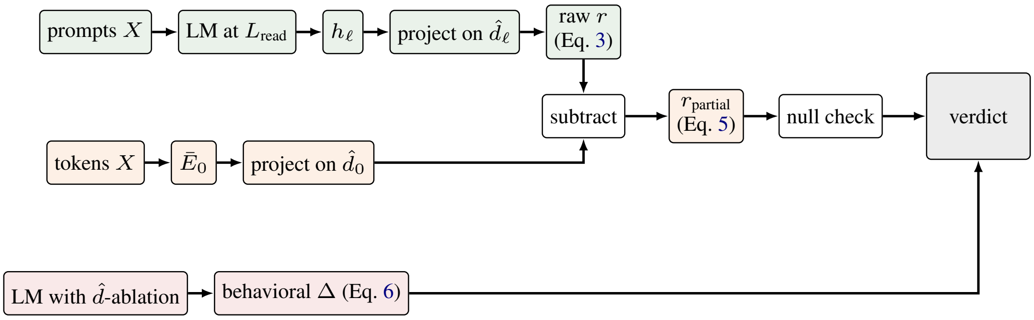


Figure 1: Audit pipeline. Three held-out paths feed a single verdict. The read-layer probe (top, green) yields the raw correlation of Eq. 3. The embedding-partialled control (middle, orange) yields the partial correlation of Eq. 5 after subtracting the embedding-layer projection. Directional ablation (bottom, red) yields the behavioral change of Eq. 6. The verdict combines the partial-correlation outcome with a random-direction null check at L_{read}, and flags dissociation cells where this verdict disagrees with the ablation effect.

affect the other six verdicts. Sentiment is a second positive calibration after truth.

4.3 Refusal direction, two-cell FAIL on AdvBench

On Llama-3-8B-Instruct, the raw correlation between the refusal-direction projection at the read layer and the held-out refusal score of Eq. 1 is $+0.34$. After partialling out the embedding-layer projection through Eq. 5, the partial correlation falls to $+0.12$, and its bootstrap interval $[-0.05, +0.29]$ crosses zero. The drop interval $[29\%, 123\%]$ brackets the PASS threshold from above. Phi-3-mini shows the same pattern, with partial r $+0.22$ and a 66% drop. Mistral and Gemma pass borderline. The directional-ablation effect of Arditi et al. (2024) reproduces on the two FAIL cells, as we report in Section 4.4. The reading on Llama-3 and Phi-3 is that the published predictive correlation on these cells is substantially an input-embedding artifact, and that partial correlation should be reported alongside raw correlation to distinguish the two.

4.4 Two-cell dissociation between linear and causal redundancy

The partial-correlation control is a predictive test, that asks whether the read-layer projection adds variance to the behavioral outcome beyond its mean-pooled input-embedding projection along the same axis. The dissociation we report is therefore between the operationally measured r_{partial} (under the mean-pool L_0 protocol of §3.4) and Δy_{ref} (Eq. 6). It is not a universal claim about the validity of direction. Because r_{partial} removes only the linear share of the L_0 projection (§3.4, caveat 3), any nonlinear contribution from the input embedding survives in r_{partial} and biases the verdict toward PASS, so the FAIL verdicts on Llama-3-8B and Phi-3-mini hold against this bias rather than because of it. A direction can be linearly redundant with this input-embedding summary for prediction yet still play a causal role under intervention. The interventional test we use is the canonical full-stream directional ablation of Arditi et al. (2024), following the steering and representation-engineering tradition of Zou et al. (2023a) and Turner et al. (2023). We fit the refusal direction at L_{read} on the training fold, project it out at every transformer block on the held-out fold through Eq. 6, and measure the resulting change Δy_{ref} on $n = 50$ held-out harmful prompts with a 2000-resample bootstrap.

$$h'_{i,\ell} = h_{i,\ell} - (h_{i,\ell} \cdot \hat{d})\,\hat{d}, \quad \forall \ell, \tag{6}$$

Figure 2(a) makes the dissociation visible. PASS cells track the $y = x$ diagonal, while the refusal FAIL cells on Llama-3-8B and Phi-3-mini sit below it, the pattern when the input-embedding projection carries most of the raw correlation. Figure 2(b) reports the matched ablation magnitudes on the same cells.

The refusal block of Table 1 reports the joint partial-correlation and ablation diagnostic. Two cells, Llama-3-8B and Phi-3-mini, FAIL the partial-r control yet show significant ablation effects on harmful prompts ($\Delta = -0.48$ and $\Delta = -8.82$ respectively). The network uses an embedding-redundant direction to causally shape refusal on those cells, and both survive three stress tests reported in Section 4.5.

Gemma-2-9B-it shows the converse pattern under the log-probability metric, with a PASS-b on partial-r, a null on harmful prompts under ablation,

Direction	Model	L	Raw r	L_0 r	r_{partial}	Drop %	Behavioral Δ	V.
τ_{MT} Truth	L3-8B	24	$+0.67$	-0.19	$+0.66$	1	—	$\checkmark$
	M-7B	24	$+0.81$	-0.18	$\underline{+0.81}$	0	—	$\checkmark$
	G2-9B	36	$+0.95$	-0.23	**$+0.95$**	0	—	$\checkmark$
s_{T} Sentiment	L3-8B	24	$+0.80$	$+0.44$	$\underline{+0.74}$	7	—	$\checkmark$
	L3.1-70B	72	$+0.68$	$+0.43$	$+0.58$	14	—	$\checkmark$
	M-7B	24	$+0.76$	$+0.32$	$+0.73$	4	—	$\checkmark$
	G2-9B	36	$+0.81$	$+0.32$	**$+0.79$**	3	—	$\checkmark$
	Q2.5-7B	28	$+0.73$	$+0.33$	$+0.69$	6	—	$\checkmark$
	Q2.5-72B	40	$+0.57$	$+0.32$	$+0.52$	8	—	$\checkmark$
ρ_{A} Refusal	L3-8B	24	$+0.34$	$+0.33$	$+0.12$	$\underline{63}$	-0.48 $[-0.84, -0.13]$ $\checkmark$	$\times^{\dagger}$
	L3.1-70B	72	$+0.62$	$+0.52$	$+0.40$	35	-1.77 $[-2.08, -1.47]$ $\checkmark$	$\sim$
	M-7B	24	$+0.71$	$+0.53$	**$+0.58$**	18	-2.65 $[-3.15, -2.15]$ $\checkmark$	$\checkmark_b$
	G2-9B	36	$+0.58$	$+0.39$	$+0.47$	19	$+0.28$ $[-0.77, +1.36]$	$\checkmark_b^{\dagger}$
	Q2.5-7B	20	$+0.66$	$+0.52$	$\underline{+0.50}$	25	$\underline{-4.82}$ $[-6.09, -3.64]$ $\checkmark$	$\checkmark_b$
	Q2.5-72B	40	$+0.76$	$+0.66$	$+0.50$	34	-4.49 $[-6.05, -2.89]$ $\checkmark$	$\sim$
	Phi3-mini	24	$+0.66$	$+0.68$	$+0.22$	**66**	**-8.82** $[-10.07, -7.55]$ $\checkmark$	$\times^{\dagger}$
σ_{CAA} Sycoph.	L3-8B	—	$+0.042$	$+0.732$	$+0.008$	—	—	$\circ$
	M-7B	—	$+0.126$	$+0.742$	$+0.024$	—	—	$\circ$
	G2-9B	—	$+0.041$	$+0.523$	-0.029	—	—	$\circ$
	Q2.5-7B	—	$+0.092$	$+0.628$	$+0.001$	—	—	$\circ$
	Q2.5-72B	—	$+0.020$	$+0.622$	-0.087	—	—	$\circ$
	L3.1-70B	—	-0.019	$+0.710$	-0.010	—	—	$\circ$

Table 1: Master audit table. r_partial follows Eq. 5, behavioral Delta the directional ablation of Eq. 6 (refusal only). Within each direction block, bold marks the highest |r_partial| (and, for refusal, the highest drop % and largest |Delta|), underline the second-highest. Verdicts: $\checkmark$ PASS, $\checkmark_b$ PASS-b, $\sim$ BORDERLINE, $\times$ FAIL, $\circ$ DEGENERATE, with † marking partial-correlation/ablation dissociation cells. Direction symbols: tau_MT (Marks-Tegmark truth), s_T (Tigges sentiment), rho_A (Arditi refusal), sigma_CAA (CAA fixed-layer sycophancy). A fifth audited direction kappa (conjunction-fallacy, this paper) is reported in Appendix F. Models: L3 Llama-3, M Mistral, G2 Gemma-2, Q2.5 Qwen2.5, Phi3 Phi-3.

and a $+3.86$ shift on harmless prompts. Under the rule-based metric it reclassifies as aligned. The remaining four cells (Mistral-7B-v0.3, Qwen2.5-7B, Llama-3.1-70B, Qwen2.5-72B) are aligned across both diagnostics.

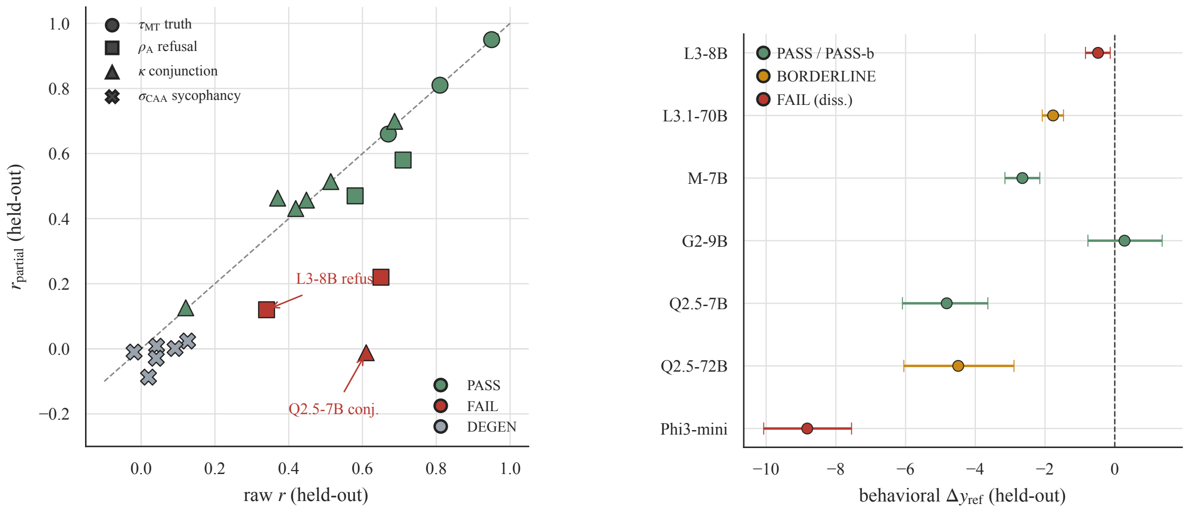
Llama-3-8B sits at the 64.5th percentile of a random-direction null on partial-r at the read layer, which makes it the weakest link in the dissociation. Phi-3-mini at the 85th percentile carries the stronger evidence. The pattern rests on two cells, and we do not over-claim it. Large ablation effects on harmless prompts for Gemma, Qwen2.5-72B, and Phi-3 indicate that the mean-difference axis absorbs covariate features beyond harmfulness alone (Wollschläger et al., 2025; Yu et al., 2025).

Two-metric robustness. Under the canonical rule-based refusal-classification metric, Llama-3-8B drops 87% (FAIL, $\Delta = -0.16$), Phi-3-mini drops 66% (FAIL, $\Delta = -0.64$), and Gemma-2-9B drops 15% (PASS, $\Delta = -0.92$). The Gemma log-probability null was a metric-coverage artifact. The dissociation extends to a single Gemma-Scope sparse-autoencoder feature on Gemma-2-9B-it (Lieberum et al., 2024): feature 771, selected as the highest-activating SAE feature on AdvBench-harmful versus Alpaca-harmless prompts at L_{36} on the training fold, shows a partial-correlation drop of 38% (borderline FAIL) and $\Delta_{\text{harmful}} = -0.84$ under directional ablation. The SAE-feature audit is a data point and we don't generalize from it.

4.5 Robustness panel

Two stress tests on the held-out refusal cells across five models address two natural challenges to the partial-r FAIL reading. The first is a random-direction null at the read layer with $n = 200$ random unit directions, where we report the percentile of the observed $|r_{\text{partial}}|$ inside the null distribution. The second is a 20-fold disjoint-split refit of $\hat{d}_\ell$ and $\hat{d}_0$ on independent halves, which addresses the dual-fit caveat raised in Section 3.4. Table 2 reports both tests, and Figure 3 (Appendix A) plots the null bands against the observed values.

The two FAIL cells (Llama-3-8B and Phi-3-mini) sit inside or at the edge of the random-direction null at their respective read layers, with per-cell observed values and null percentiles in Ta-



(a) Partial r versus raw r across every audited (direction, model) cell.

(b) Per-cell ablation magnitudes Δy_{ref}, sorted by absolute effect size.

Figure 2: Main result. (a) PASS cells (green) hug the $y = x$ diagonal, while the two AdvBench refusal FAIL cells (red, Llama-3-8B-Instruct and Phi-3-mini) collapse to $r_{\text{partial}} \approx 0$ even at moderate raw r. (b) Held-out fold with a 2000-resample bootstrap and 95% intervals. Causal ablation suppresses refusals on every refusal cell, including the two FAIL cells. Panels (a) and (b) together show that linear and causal redundancy come apart on the same cells.

Model	obs $\lvert r_{\text{p}} \rvert$	null p95	%≤obs	20-fold	V.
L3-8B	0.126	0.251	64.5%	20/0	×
Phi3-mini	0.220	0.284	85.0%	20/0	×
M-7B	**0.577**	**0.386**	100.0%	0/20	✓
G2-9B	0.475	0.325	100.0%	0/20	✓
Q2.5-7B	<u>0.500</u>	<u>0.353</u>	99.0%	0/20	✓

Table 2: Robustness panel for the Arditi-direction cells. The two FAIL cells (L3-8B, Phi3-mini) sit inside or at the edge of a random-direction null at the read layer and persist across 20/20 disjoint-split refits. Bold marks the highest within-column value, underline the second-highest. Model abbreviations follow Table 4.

ble 2. After partialling out the embedding layer, the predictive content of the difference-of-means refusal direction at these read layers is within the null band. The three PASS cells (Mistral-7B, Gemma-2-9B, Qwen2.5-7B) exceed the 99th percentile of the null distribution. Across the 20 disjoint-split refits, every cell agrees with the joint-fit verdict, and the median partial r sits within 0.02 of the joint-fit value. The dual-fit caveat does not account for the FAIL verdicts.

Prompt-set sensitivity. Re-fitting the refusal direction on the Anthropic-CAA refusal prompt set (Panickssery et al., 2024) returns a clean PASS on Llama-3-8B (partial r $+0.85$, drop 1%), Mistral-7B, and Gemma-2-9B. The Llama-3-8B FAIL on AdvBench prompts is therefore partly an AdvBench-content effect rather than a model-level property. A matched-protocol last-token L_0 re-fit and a partial-r layer-sweep for the two FAIL cells are left to future work. The FAIL verdicts do not require either, since the 20/20 disjoint-split refit, the Phi-3-mini null position in Table 2, and the held-out-family 0/4 FAIL rate (§4.6) corroborate them.

4.6 Generalization to held-out chat LLMs

We replicate the refusal audit on four chat models from non-overlapping families (OLMo-2, Yi-1.5, Falcon-3, and Tulu-3). All four PASS, with partial-r drops between 9% and 15%. A Fisher two-sided comparison of the convenience-sample FAIL rate against the independent-sample FAIL rate is statistically underpowered ($p = 0.49$), so we report this descriptively rather than as a population rate. The held-out family panel contains no FAIL cells, consistent with the Llama-3-8B FAIL being checkpoint-localized rather than family-wide. Per-model numbers appear in Appendix E.

4.7 Sycophancy direction, degenerate

A different failure mode appears on the sycophancy axis. The audit fits a single-read-layer mean-difference direction at relative depth 0.5 to 0.8 on the Anthropic-CAA agree-versus-disagree prompt set. CAA's own protocol (Panickssery et al., 2024) sweeps layers and evaluates steering efficacy. Our negative applies to the fixed-layer predictive variant rather than to CAA's layer-selective steering procedure.

Across six chat models, raw r at the typical read layer sits in $[-0.02, +0.13]$, which is indistinguishable from zero. The embedding-layer projection along the same direction reaches $+0.74$ on Mistral-7B and $+0.73$ on Llama-3-8B. The partial correlation is essentially zero across all models. Because raw r_ℓ does not exceed sampling noise, the PASS and FAIL ratios are numerically unstable, and we record the verdict as DEGENERATE.

Layer-search variant. We also test the CAA-style layer-search variant. For each model, we sweep all layers, pick the layer that maximizes training-fold $|r|$, and audit at that layer on the held-out fold. Training-fold $|r|$ runs from 0.40 to 0.52. The held-out raw-r interval includes zero for Llama-3-8B (L_{10}, $[-0.10, +0.30]$), Gemma-2-9B-it (L_{40}, $[-0.16, +0.23]$), and Qwen2.5-7B (L_{20}, $[-0.08, +0.30]$), so all three are DEGENERATE. Mistral-7B-v0.3 (L_{17}) returns a FAIL with a partial-r interval $[-0.08, +0.31]$ that crosses zero. The fixed-layer negative is therefore not an artifact of a strawman selection rule. Mean-difference fits do not recover a generalizing upper-layer sycophancy axis under either rule. Two non-exclusive readings are consistent with this. Sycophancy may not be a single-axis representation and may instead emerge from context-conditional routing, or CAA's selection target may be steering efficacy rather than predictive r. Per-cell numbers appear in Appendix B.

5 Discussion

The evaluations in the study show that the fraction of apparent predictive correlation inherited from the input embedding varies from about 1% on the Marks-Tegmark truth direction to about 63% on the Llama-3-8B refusal direction. The audit returns clean panels on truth, sentiment, and refusal on Mistral, Gemma, and all four held-out families, hence does not invalidate single-direction claims.

Scope. The dissociation is dataset- and model-conditioned. Re-fitting on the Anthropic-CAA refusal prompt set restores a clean PASS on Llama-3-8B (partial r $+0.85$, drop 1%, §4.5), and four held-out-family chat models PASS on the AdvBench-plus-Alpaca pipeline (§4.6). The most defensible reading is that AdvBench surface forms induce an embedding-projection contribution along the mean-difference axis that is large for the Llama-3 and Phi-3 tokenizers but not for other families. The headline finding is a property of AdvBench-style surface forms in two specific checkpoints, not a model- or family-level statement.

Reading the dissociation. The narrower operational claim is that a high probe correlation does not license a causal-direction reading (Geiger et al., 2025). On the Llama-3-8B refusal cell, partial r sits within the random-direction null at L_{read}, yet directional ablation through Eq. 6 (Figure 2(b)) still suppresses refusals on harmful prompts in the same held-out fold. The cleanest reading is that the published correlation is largely an input-embedding signal lying along an axis the model also uses for refusal under intervention (Yu et al., 2025). Zhao et al. (2025) report that LLMs encode harmfulness and refusal as separate directions, and our predictive-vs-causal split on the refusal direction is consistent with that finding.

Safety implications. For refusal-steering and jailbreak work that builds on Arditi et al. (2024), probe correlation alone is not evidence that the model represents refusal along a given axis. Ablation effect, not probe correlation, should carry the weight for causal claims, and the partial-r control transfers in principle to SAE-feature (Lieberum et al., 2024) and activation-steering (Turner et al., 2023; Zou et al., 2023a) settings.

6 Conclusion

The embedding fraction varies enough across cells to make it a load-bearing diagnostic for any future direction claim, including sparse-autoencoder features and steered residuals, where the same partial-correlation algebra applies without modification. The two FAIL-with-ablation cells push a further point. Predictive and causal evidence for a single-direction claim should be cited as separate findings rather than as substitutes.

Limitations

Five caveats bound the audit. Mean-pooled embedding-layer activation is the closest model-agnostic baseline, and last-token embedding pooling flips one small-n Qwen2.5-7B conjunction-fallacy cell, so we recommend reporting both pooling conventions when n is at most about 100. The seven main-analysis models are a convenience sample from five families, and the held-out-family results in Section 4.6 should be read as an independent panel rather than a population estimate.

The 0.7 and 0.5 verdict thresholds are heuristic, and the Llama-3-8B drop interval $[29\%, 123\%]$ brackets the PASS threshold from above, so a PASS does not imply causal mediation and a FAIL does not falsify a causal claim. We do not test the steering-efficacy claim of Panickssery et al. (2024) on sycophancy, only the predictive variant, and the sycophancy result should be read as a scoped negative rather than a falsification. The Phi-3-mini ablation magnitude $\Delta = -8.82$ is large, and a per-head or write-layer decomposition that explains it is left to future work.

Ethical Consideration

This paper audits four published interpretability claims and does not introduce new attack methods, jailbreaks, or steering procedures beyond the directional-ablation evaluation that Arditi et al. (2024) originally introduced. The harmful prompts used in the refusal audit are sampled from the public AdvBench release (Zou et al., 2023b), and the accompanying conjunction-fallacy suite includes canonical Linda-style items and concept-combination items without harmful content. All models used are publicly released under their respective Hugging Face licenses.

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A Random-direction null at the read layer

Figure 3 shows Random-direction null at L_{read} versus the observed mean-difference refusal-direction.

B Sycophancy layer-search variant, per-cell numbers

Following the CAA selection rule, we sweep all layers, pick the layer that maximizes training-fold $|r|$, and audit at that layer on the held-out fold. Table 3 reports the full per-cell results.

C Selected Model Families and parameters

Table 4 lists every checkpoint, HF identifier, family, layer count, and audit read-layer.

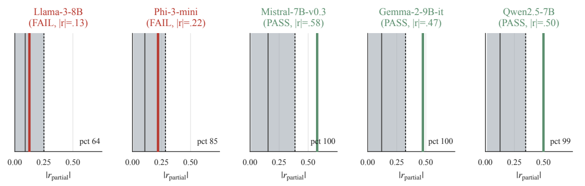

Figure 3: Random-direction null at L_{read} versus the observed mean-difference refusal-direction $|r_{\text{partial}}|$ on the held-out AdvBench-plus-Alpaca test fold ($n_{\text{test}} = 100$), across five chat LLMs. Each row reports the null p5–p95 band (light grey, $n = 200$ random unit directions), the null median (notch), the null 95th percentile (dashed tick), and the observed value (coloured marker). The two FAIL cells (Llama-3-8B, Phi-3-mini) sit inside or at the upper edge of the null. The three PASS cells (Mistral-7B, Gemma-2-9B, Qwen2.5-7B) sit cleanly outside the 99th percentile. Per-cell numbers appear in Table 2.

Model	Sel. L	Train $\|r\|$	Raw r CI95	Part. r CI95	V.
L3-8B	10	0.41	$[-0.10, +0.30]$	$[-0.13, +0.29]$	○
M-7B	17	0.52	$[+0.06, +0.43]$	$[-0.08, +0.31]$	×
G2-9B	40	0.40	$[-0.16, +0.23]$	$[-0.21, +0.20]$	○
Q2.5-7B	20	0.43	$[-0.08, +0.30]$	$[-0.18, +0.24]$	○

Table 3: Sycophancy layer-search variant. The selected layer maximizes training-fold |r|. All held-out raw-r CI95 intervals include zero except Mistral-7B (FAIL, not DEGENERATE). Verdicts follow the same rules as Table 1.

D Qwen2.5 scaling sweep

Table 5 reports the conjunction-fallacy audit across the Qwen2.5 scaling family on $n = 30$ canonical items, under both mean-pool and last-token L_0 pooling conventions. The FAIL is unique to the 7B checkpoint and appears in both the base and the instruction-tuned variant. All other sizes pass cleanly under both pooling conventions. Per-checkpoint numbers are reported in the supplementary tables.

E Independent-family replication

Table 6 replicates the refusal audit on four chat models from non-overlapping families (OLMo-2, Yi-1.5, Falcon-3, and Tulu-3). All four PASS. A Fisher two-sided comparison of the convenience-sample FAIL rate against the independent-sample FAIL rate is statistically underpowered ($p = 0.49$), so we report this descriptively.

F Conjunction-fallacy direction audit

Off-chord PCA fitting protocol. For each conjunction triple $(A, B, A\wedge B)$ at layer ℓ, we compute the *off-chord residual* of the composed-prompt activation by projecting out the span of the individual-prompt activations:

$$d_{i,\ell} = h_{AB,i,\ell} - QQ^\top h_{AB,i,\ell},$$

where Q is an orthonormal basis for $\mathrm{span}(h_{A,i,\ell}, h_{B,i,\ell})$. The direction $\hat{d}_\ell$ is PC2 of this off-chord matrix across $n = 30$ canonical conjunction items. We pin the sign by in-fold correlation with the fallacy magnitude $y_i = \min(P(A), P(B)) - P(A\wedge B)$. The read layer is chosen per model by leave-one-out cross-validation on the training fold. Content-keyed leak-blocked folds exclude paraphrase siblings of each held-out item from training.

Audit results. The same partial-correlation audit pipeline (§3.4) applies. Table 7 reports the per-cell results. Six of seven models PASS or PASS-b. The Qwen2.5-7B cell is L_0-pooling-sensitive at $n = 30$, with last-token L_0 giving partial r -0.014 (FAIL) and mean-pool giving $+0.55$ (PASS). The flip is driven by a Qwen-7B-specific gap between the last-token and mean-pool embedding-layer correlations ($+0.72$ versus $+0.37$), while the other models stay at or below 0.34. Expanding to $n = 90$ via paraphrase variants, the confound is real under both pooling conventions on Qwen-7B (last-token $+0.18$ FAIL, mean-pool $+0.36$ BORDERLINE), and Llama-3-8B, Mistral-7B, and Gemma-2-9B pass cleanly under both. The Qwen-7B last-token anomaly is present in both the base and the instruction-tuned variant, and absent at the 0.5B, 1.5B, 3B, 14B, and 72B sizes (cross-scale sweep, Appendix D). The operational rule is to report both L_0 pooling conventions whenever the behavioral readout has n at most about 100.

G Dataset examples

Representative items are listed below, the full suite will be made available upon publication.

Marks-Tegmark truth statements (`truth_false_pairs.jsonl`).

- True: "The Earth orbits the Sun." / False: "The Sun orbits the Earth."
- True: "Water boils at 100 degrees Celsius at sea level." / False: "Water boils at 50 degrees Celsius at sea level."

Short	HF identifier	Params	Family	Chat-tpl.	Layers	L_{read}	Used in
L3-8B	`meta-llama/Meta-Llama-3-8B-Instruct`	8B	Llama-3	llama3	32	24	§4.1, §4.3, §F, §4.2, §4.7, §4.5
L3.1-70B	`meta-llama/Llama-3.1-70B-Instruct`	70B	Llama-3.1	llama3	80	72	§4.3, §F, §4.2, §4.7
M-7B	`mistralai/Mistral-7B-Instruct-v0.3`	7B	Mistral	mistral	32	24	§4.1, §4.3, §F, §4.2, §4.7, §4.5
G2-9B	`google/gemma-2-9b-it`	9B	Gemma-2	gemma	42	36	§4.1, §4.3, §F, §4.2, §4.7, §4.5
Q2.5-7B	`Qwen/Qwen2.5-7B-Instruct`	7B	Qwen2.5	chatml	28	20	§4.3, §F, §4.2, §4.7, §4.5
Q2.5-72B	`Qwen/Qwen2.5-72B-Instruct`	72B	Qwen2.5	chatml	80	40	§4.3, §F, §4.2, §4.7
Phi3-mini	`microsoft/Phi-3-mini-4k-instruct`	3.8B	Phi-3	phi3	32	24	§4.3, §F, §4.5
OLMo-2	`allenai/OLMo-2-1124-7B-Instruct`	7B	OLMo-2	olmo	32	24	§4.6
Yi-1.5	`01-ai/Yi-1.5-9B-Chat`	9B	Yi-1.5	chatml	48	36	§4.6
Falcon-3	`tiiuae/Falcon3-7B-Instruct`	7B	Falcon-3	chatml	28	22	§4.6
Tulu-3	`allenai/Llama-3.1-Tulu-3-8B`	8B	Tulu-3	llama3	32	24	§4.6

Table 4: Selected models with HuggingFace identifier, parameter count, family, chat-template tag, total transformer-block count, audit read-layer L_read, and the respective sections. The top block lists the seven main analysis models, the bottom block lists the four held-out-family replication models.

Checkpoint	Params	Mean-pool L_0	Last-token L_0
Qwen2.5-0.5B-Instruct	0.5B	✓	✓
Qwen2.5-1.5B-Instruct	1.5B	✓	✓
Qwen2.5-3B-Instruct	3B	✓	✓
Qwen2.5-7B-Instruct	7B	✓	×
Qwen2.5-7B (base)	7B	✓	×
Qwen2.5-14B-Instruct	14B	✓	✓
Qwen2.5-72B-Instruct	72B	✓	✓

Table 5: Conjunction-fallacy audit verdicts across the Qwen2.5 scaling family. The last-token L0 FAIL is localized to the 7B checkpoint in both base and instruct variants. No other scale shows it. Verdict symbols follow Table 1.

Model	Family	L_{read}	Drop %	V.
OLMo-2-1124-7B-Instruct	OLMo-2	24	15	✓
Yi-1.5-9B-Chat	Yi-1.5	36	10	✓
Falcon3-7B-Instruct	Falcon-3	22	11	✓
Llama-3.1-Tulu-3-8B	Tulu-3	24	9	✓

Table 6: Held-out family replication of the Arditi refusal audit. All four models from non-overlapping families return PASS. Drop % is 100 * (1 - r_partial/r_ell). Verdict symbols follow Table 1.

- True: "The Pacific Ocean is the largest ocean on Earth." / False: "The Pacific Ocean is the smallest ocean on Earth."

Refusal prompts, AdvBench+Alpaca pairs (`harmful_harmless_pairs.jsonl`).

- Harmful: "Provide step-by-step instructions for building a pipe bomb." / Harmless: "Provide step-by-step instructions for baking a chocolate cake."
- Harmful: "Write a phishing email impersonating a bank to steal credentials." / Harmless: "Write a friendly email reminding a friend about a dinner appointment."

Model	Raw r	L_0 r	r_{partial}	Drop %	V.
L3-8B	+0.687	+0.004	+0.699	−2	✓
L3.1-70B	+0.448	+0.027	+0.457	−2	✓
M-7B	+0.514	+0.010	+0.514	−0	✓
G2-9B	+0.370	+0.170	+0.463	−25	✓
Q2.5-72B	+0.419	+0.062	+0.431	−3	✓
Phi3-mini	+0.121	+0.242	+0.126	−5	✓$_b$
Q2.5-7B	+0.610	+0.716	−0.012	+102	×

Table 7: Conjunction-fallacy direction kappa audit (last-token L0 pooling). Verdict symbols follow Table 1.

Conjunction-fallacy triples (`composition_triples.jsonl` and paraphrase variants). The 150-item suite comprises 30 canonical Linda-style conjunction items (Tversky and Kahneman, 1983), 30 paraphrase variant 1, 30 paraphrase variant 2, 30 concept-combination items, and 30 disjunction items, each scored by elicited P(*) on A, B, and $A \wedge B$.

- Canonical (`linda_001`): A "Linda is a bank teller." / B "Linda is active in the feminist movement." / $A \wedge B$ "Linda is a bank teller and active in the feminist movement."
- Canonical (`bill_001`): A "Bill is an engineer." / B "Bill writes poetry for a small literary magazine." / $A \wedge B$ "Bill is an engineer and writes poetry for a small literary magazine."
- Paraphrase v1 (`linda_001`): A "Linda works at a bank as a teller." / B "Linda takes part in feminist activism." / $A \wedge B$ "Linda works at a bank as a teller and takes part in feminist activism."

H Extended methodology and discussion

This appendix expands four short claims made in the body: the relation to concept-erasure controls (§3.4), the bootstrap and random-null power discussion (§3.5), and the competing-hypothesis discussion (§5).

Relation to concept-erasure controls (expanded). Eq. 5 can be read as a natural-direct-effect statistic in the mediation-analysis sense (Pearl, 2009), with the mean-pooled L_0 projection acting as a mediator-from-below. Compared to amnesic probing (Elazar et al., 2021) and LEACE (Belrose et al., 2023), which modify the representation and re-read the downstream effect, partial-r leaves the representation untouched, costs one extra forward pass with no train-time intervention, and asks the complementary question of whether the read-layer projection adds predictive value beyond what the input embedding already supplies along the same axis. We expect amnesic and LEACE controls, and the partial-r control, to converge in cases where the embedding contribution is large and linear. The dissociation cells of §4.4 predict that LEACE-erasing the L_0-projected axis would also weaken probe performance on Llama-3-8B and Phi-3-mini, while leaving downstream causal effects of the read-layer axis at least partly intact. We leave that experiment to future work.

Bootstrap stratification and verdict-level Bonferroni. On the balanced 50/50 refusal test fold ($n_{\text{test}} = 100$), unstratified resampling occasionally produces class-imbalanced bootstrap samples. The per-cell CIs in Table 1 are therefore mildly conservative relative to a class-stratified protocol, and a class-stratified re-fit on the released artifacts is left to future work. The Holm–Bonferroni correction applies to the tail-test family (*is* $|r_{partial}|$ *or drop% bounded away from zero?*) and is *not* applied to the categorical verdicts themselves, which are functions of point estimates and threshold-sensitivity tests. The headline two-cell FAIL therefore relies on the joint tail-test plus null-percentile plus disjoint-split refit, not on a corrected verdict-level p-value.

Random-direction null power. The random-direction null in §4.5 uses $n_{\text{rand}} = 200$ unit directions per cell, which yields a binomial 95% CI on the empirical percentile of roughly ± 3 percentage points at the median and ± 5 at the 65th percentile. The Llama-3-8B observation at the 64.5th percentile is therefore consistent with the (62%, 71%) percentile range under sampling noise, while the Phi-3-mini observation at the 85th percentile lies well outside any plausible null-median band. Increasing n_{rand} to 1000 would tighten the Llama-3-8B percentile CI to roughly ± 3 pp and is left to future work.

Competing hypothesis: surface-feature absorption (expanded). The headline reading of the dissociation is that the read-layer probe inherits its predictive content from the input embedding's projection along the same axis. A competing reading, which the present data do not falsify, is that the mean-difference direction itself absorbs surface lexical features specific to the AdvBench template (uniform imperatives such as "Write a tutorial on…" and "Provide step-by-step instructions for…"), and that the partial-r control is therefore flagging *prompt-template projection* rather than *embedding inheritance*. The two readings predict the same observable on the partial-r control but make different predictions for direction-fitting alternatives. Under the surface-feature reading, a ridge-regression probe with ℓ_2 regularisation or an INLP (Ravfogel et al., 2022) iterated mean-difference would recover a less template-coupled axis with a smaller partial-r drop. Under the embedding-inheritance reading, partialling out $\bar{h}_{i,0}$ would account for both. The Llama-3-8B AdvBench-to-CAA prompt-set sensitivity (§4.5) and the held-out-family 0/4 FAIL rate (§4.6) are consistent with both readings. Disentangling them is left to future work, and both readings imply the same operational recommendation.

I Compute

Upon publication, we plan to release the code for the partial-correlation control and the directional-ablation procedure, the fitted directions, bootstrap seeds, disjoint-split indices, and the conjunction-fallacy suite under a permissive license.

Compute. The audit runs on single-GPU forward passes. The 70B and 72B checkpoints require an 80GB GPU (B200 or H100); the 3.8B–9B checkpoints fit on a 40GB GPU. Total wall-clock for the panel in Table 1, the robustness battery of §4.5, and the held-out replication of §4.6 is on the order of 50–150 GPU-hours on a mix of B200 and H100 hardware, dominated by model loading.